PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

Introduce your team and the domain you have chosen

- Team Name – **Aristella**
- Team Members – **Dhiren Vasudeva, Vani Karre, Annareddy Pardhav Reddy, Dhanushiya S**
- Selected Domain – **#2**
- Problem Statement – **SolarSense**
  Weather-aware energy planning for homes with solar and battery backup

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DOMAIN PROBLEM IN OUR WORDS

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Explain the problem in your domain simply, in your own words

“My battery says 62%. Can I run the AC tonight?”

The answer depends on three things, not the percentage alone:

Usable battery energy
After minimum charge, battery health and losses

Essential loads
Fridge, router and lights come first

Expected generation
Tomorrow's sunshine changes tonight's margin

Battery readings and energy charts can still leave households unsure which appliance choices fit.

What exactly needs to be solved?

Turn readings, priorities and the forecast into a clear answer: what fits, for how long, and what to keep.

Why is it important right now?

Rooftop solar is growing quickly in India, so more households face this decision every evening.

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WHO IS AFFECTED?

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Target users, stakeholders, beneficiaries

Primary users / target group

Households with rooftop solar, compatible battery backup and flexible appliance use.

Secondary stakeholders

Installers, service teams, and family members sharing the system.

Who benefits if the problem is solved?

Households making clearer decisions, and installers supporting them.

Later expansion: small shops with battery backup. **Scope:** not for systems without a battery, or for medical equipment.

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CURRENT SITUATION & CONTEXT

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Existing scenario, background and how things work today

Existing scenario and background

Homes with solar and battery backup store daytime energy for evenings and outages. Monitoring apps report charge, load and generation; households still decide what to run.

Relevant data or facts

50.06 lakh

households benefiting from rooftop solar under PM Surya Ghar (PIB, 4 Aug 2026). Not all have battery backup.

Current workflow, step by step *Illustrative workflow, to validate*

1. **Check charge** — Look at the battery reading
2. **Estimate runtime** — Judge what it can run
3. **Use appliances** — Essentials and optional together
4. **Adjust** — After an unexpected shortfall, if one occurs

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Main difficulties and bottlenecks	Root causes behind them	Consequences of leaving them unsolved
Unclear usable energy	Usable energy depends on minimum charge, health and losses	Appliance runtime is hard to judge
Competing appliance priorities	Essential and optional loads share one battery	Optional use can eat into energy needed overnight
Uncertain generation and demand	Tomorrow's sunshine and household use both vary	Shortfalls are noticed late instead of planned for

SolarSense proposes to address these challenges for battery-backed households.

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EXISTING SOLUTIONS / APPROACHES

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What is already being done, and where it falls short

Existing methods, systems or alternatives	What they offer	Limitations of each, for our users
Monitoring and insights e.g. Luminous app	System status and insights	Readings may still need interpreting into appliance choices
Weather preparation Tesla Storm Watch	Charges the battery ahead of severe-weather alerts	Event-based; for Powerwall owners
AI assistance Enphase AI assistant (Aug 2026)	Answers questions about the owner's system	Availability varies by market and system
Forecast-based optimisation EMHASS, Forecast.Solar	**EMHASS:** forecast-based scheduling **Forecast.Solar:** solar-generation forecasts	**EMHASS** requires integration and configuration **Forecast.Solar** needs a separate household planner

The gap they leave open — SolarSense's proposed approach

SolarSense budgets for household essentials, then explains which optional appliances fit—and for how long.

PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

Insight: the opportunity is a decision layer, not new hardware. Manual entry is a plausible starting point; automated integrations would need model-specific testing.

Hardware Requirements

- Existing compatible solar-battery system
- Battery specifications and current charge
- Optional later: system telemetry

Software Requirements

- Weather forecast
- Physical solar model
- Appliance schedules
- Constrained energy-budget planner

Other resources (Including Human Resources)

- Mentor with energy-systems expertise
- Installer guidance on real systems
- Households for future validation

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INITIAL DIRECTION & EXPECTED OUTCOME

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What Is Your Expected Outcome?

What your team intends to do

Build SolarSense: allocate energy to essentials first in the plan, then show which appliances fit.

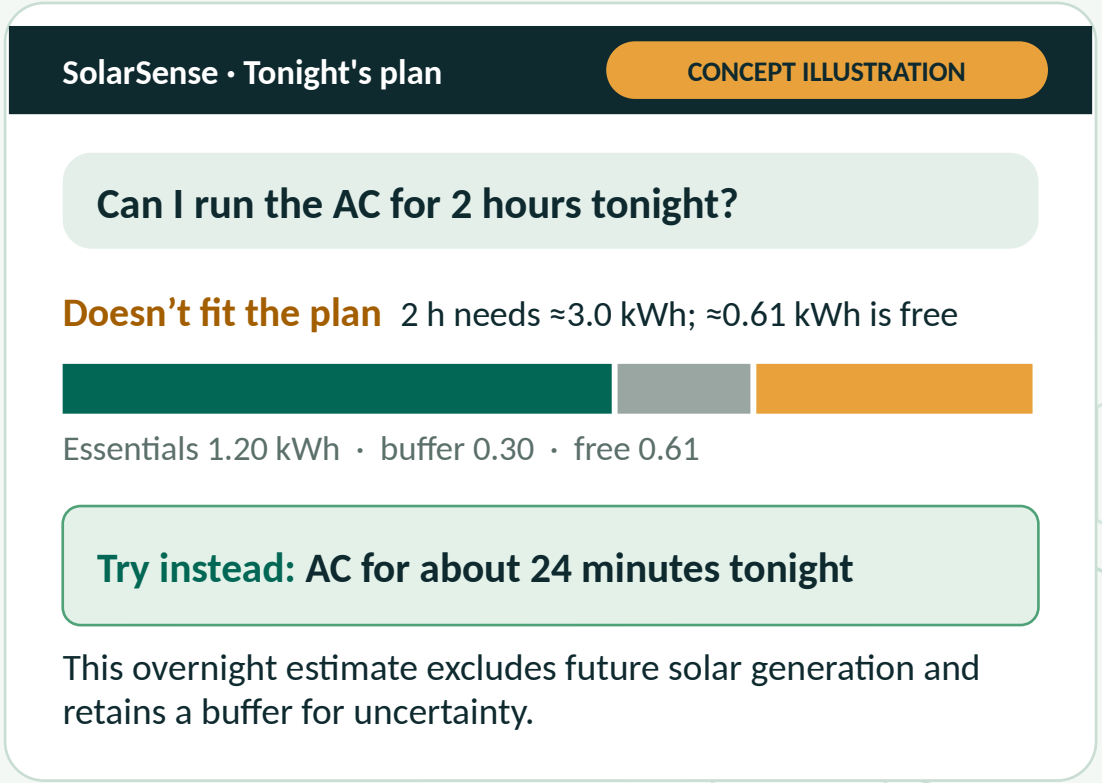
Data + priorities + forecast → Energy budget → Appliance advice

What you want to improve

- Clearer evening appliance decisions
- Essentials allocated first in the plan
- Earlier preparation for low-sun days

What successful improvement should you achieve

In future evaluation: compare forecast accuracy against simple baselines, and assess whether the resulting advice helps users make better decisions.



Illustrative scenario: 5 kWh battery, 90% health, 62% charge, 10% minimum, 90% efficiency; AC ≈1.5 kW. Power/startup compatibility needs a separate check.